



SCHOOL OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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ON

“IOT Based Sign Language Glove Using Arduino”

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DECLARATION

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We, AMRUTH B S R24EJ015, ROHAN SAI GORUMUTCHU R24EJ042, RUJULA N R24EJ111, K S BHARGAW SAI R24EJ050, students of B.Tech, belonging to the School of Computer science and technology, REVA University, declare that this IoT Report, "Sign Language Recognition Glove Using Flex Sensors," is the result of work done by us under the supervision Dr. Sayantam Sarkar, Professor, School of ECE, REVA University, Bengaluru.

We are submitting this Report in partial fulfillment of the requirements for the completion of the IoT and Applications course in I-Semester of Bachelor of Technology in Electronics and Communication Engineering by REVA University, Bengaluru, during the academic year 2024-2025.

(Signature of the Students)

January 2025

Certified that this work submitted by AMRUTH B S R24EJ015, ROHAN SAI GORUMUTCHU R24EJ042, RUJULA N R24EJ111, K S BHARGAW SAI R24EJ050 has been carried out under our guidance and the declaration made by the candidates is true to the best of our knowledge.

Signature of Faculty

Date: January 2025

Signature of Director

Date: January 2025

Official Seal of the School

Name of the Examiner with affiliation

Signature with Date

2.

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ABSTRACT

The IoT-based Sign Language Glove is an innovative project designed to help individuals with speech and hearing impairments communicate more effectively. In many situations, people who rely on sign language face difficulties due to the lack of interpreters or general public unfamiliarity with sign language. This system translates sign language gestures into text displayed on an LCD screen, making communication seamless and efficient.

The glove is equipped with multiple flex sensors placed along the fingers, which detect bending movements corresponding to different gestures. These sensors generate variable resistance values, which are processed using an Arduino Uno R3. The microcontroller then maps the input signals to predefined gestures and displays the corresponding text on a 16 x 2 LCD screen via an I2C LCD module, providing real-time feedback.

Unlike vision-based systems, which rely on cameras and image processing algorithms, this sensor-based approach is more portable, cost-effective, and independent of environmental factors such as lighting conditions. By incorporating IoT technology, this device can be further developed to integrate with wireless networks, mobile applications, or audio output systems, thus significantly improving accessibility for individuals with speech and hearing impairments. This project not only helps bridge the communication gap but also promotes inclusivity by providing a simple and effective assistive tool.

Chapter-I

Introduction

1.1.Introduction

Introduction Communication is essential for human interaction, but individuals with speech and hearing impairments often face challenges in conveying their thoughts. Sign language serves as their primary mode of communication; however, not everyone understands it. This communication gap creates obstacles in daily activities, employment opportunities, and social interactions.

The IoT-based Sign Language Glove is designed to translate sign language gestures into readable text, making communication more accessible and efficient. By leveraging modern sensor technology and microcontrollers, the glove offers a compact and user-friendly solution to aid individuals with disabilities in expressing themselves effectively.

Flex sensors mounted on gloves detect finger movements, which are converted into signals by an Arduino Uno R3. The system displays the recognized gestures on a 16 x 2 LCD screen in real-time, ensuring that the user receives immediate feedback. This portable and cost-effective device significantly enhances the quality of life for individuals relying on sign language for communication.

1.2.Objectives

1. **Accessibility:** To develop a cost-effective and easy-to-use assistive device, ensuring that individuals with speech and hearing impairments can use it without requiring advanced technical knowledge.
2. **Customization:** To allow users to modify and expand the system by adding new gestures and phrases based on individual needs, ensuring flexibility and adaptability.
3. **Interactivity:** To create a responsive system that provides real-time feedback on hand gestures through an LCD screen, making communication more effective.
4. **Portability:** To design a lightweight, battery-powered glove that can be used in different environments without dependency on external power sources.
5. **Scalability:** To ensure the system can be further developed with additional sensors or IoT features, enabling remote communication via smartphones or integration with AI-driven speech synthesis.
6. **Ease of Integration:** To make the glove compatible with other devices such as mobile phones or computers, allowing for future enhancements like voice-to-text conversion and Bluetooth connectivity.

1.3 Layout

The layout consists of several key components that work together to recognize hand gestures and display corresponding text outputs. The structure includes:

1. Input Devices:

- **Flex Sensors:** Mounted on glove fingers to detect bending movements that correspond to specific gestures.
- **Voltage Divider Circuit:** Used to process signals from flex sensors and send analog values to the Arduino.

2. Processing Unit:

- **Arduino Uno R3:** The core of the system, responsible for reading sensor inputs, processing data, and sending display commands.
- **Predefined Gesture Mapping:** Arduino maps analog sensor values to predefined text outputs.

3. Output Devices:

- **16x2 I2C LCD Module:** Displays the recognized gesture text in real-time.
- **LED Indicator:** Provides additional visual feedback.

4. Power Supply:

- **9V Battery or USB Power:** Ensures portability and continuous operation of the system.

5. Connectivity (Future Scope):

- The system can be enhanced with **Bluetooth or Wi-Fi** for remote communication with mobile applications or text-to-speech modules.

Chapter–II

2.1 Literature Survey

Literature Survey Existing methods for sign language recognition rely primarily on vision-based systems, which use cameras and image processing algorithms. While effective, these systems have limitations such as high computational requirements, dependency on lighting conditions, and privacy concerns. These constraints make vision-based systems impractical for daily use, especially in low-light environments or crowded public spaces.

Sensor-based gloves offer an alternative by using flex sensors, accelerometers, or gyroscopes to detect hand movements and gestures. Unlike vision-based recognition, sensor-based gloves can work in any lighting condition and do not require external cameras. The technology behind sensor-based gloves has been explored in various research studies, demonstrating promising results in terms of accuracy and portability.

One such system utilized flex sensors integrated with microcontrollers to detect finger movements. These sensors measured resistance changes corresponding to the degree of bending, which were then processed by microcontrollers like Arduino. Data was transmitted wirelessly to a receiver module, displaying recognized gestures on an LCD or smartphone application.

However, existing systems often lacked affordability and ease of use. Many glove-based systems required complex wiring or additional hardware, making them difficult to implement in real-world scenarios. By leveraging the Arduino Uno R3, I2C LCD modules, and flex sensors, our project aims to overcome these challenges, offering a low-cost and efficient solution for gesture recognition that is both simple and effective.

Chapter-III

3.1 Methodology of Work

The methodology followed in the development of the IoT-based Sign Language Glove includes several key steps:

1. System Design and Component Selection:

- a. Identifying key hardware components such as flex sensors, an Arduino Uno R3, an I2C LCD module, and a 9V power source.
- b. Ensuring compatibility between the selected components for seamless integration.

2. Circuit Design and Implementation:

- a. Designing the voltage divider circuit for the flex sensors to enable proper analog readings.
- b. Connecting the flex sensors to the Arduino and configuring the LCD display for text output.

3. Programming and Algorithm Development:

- a. Writing the Arduino code to read sensor values, process gesture recognition, and display corresponding text on the LCD.
- b. Implementing predefined thresholds for each gesture to ensure accurate interpretation.

4. Testing and Calibration:

- a. Conducting multiple trials to analyze sensor accuracy and adjusting threshold values as needed.
- b. Debugging any issues in gesture recognition and refining the code for optimized performance.

5. Final Integration and User Testing:

- a. Assembling all components into a wearable glove format for ease of use.

- b. Testing the system with different users to ensure reliability and usability.

6. Future Enhancements:

- a. Exploring additional sensors or connectivity features such as Bluetooth for smartphone integration.
- b. Investigating potential integration with AI models for dynamic gesture recognition

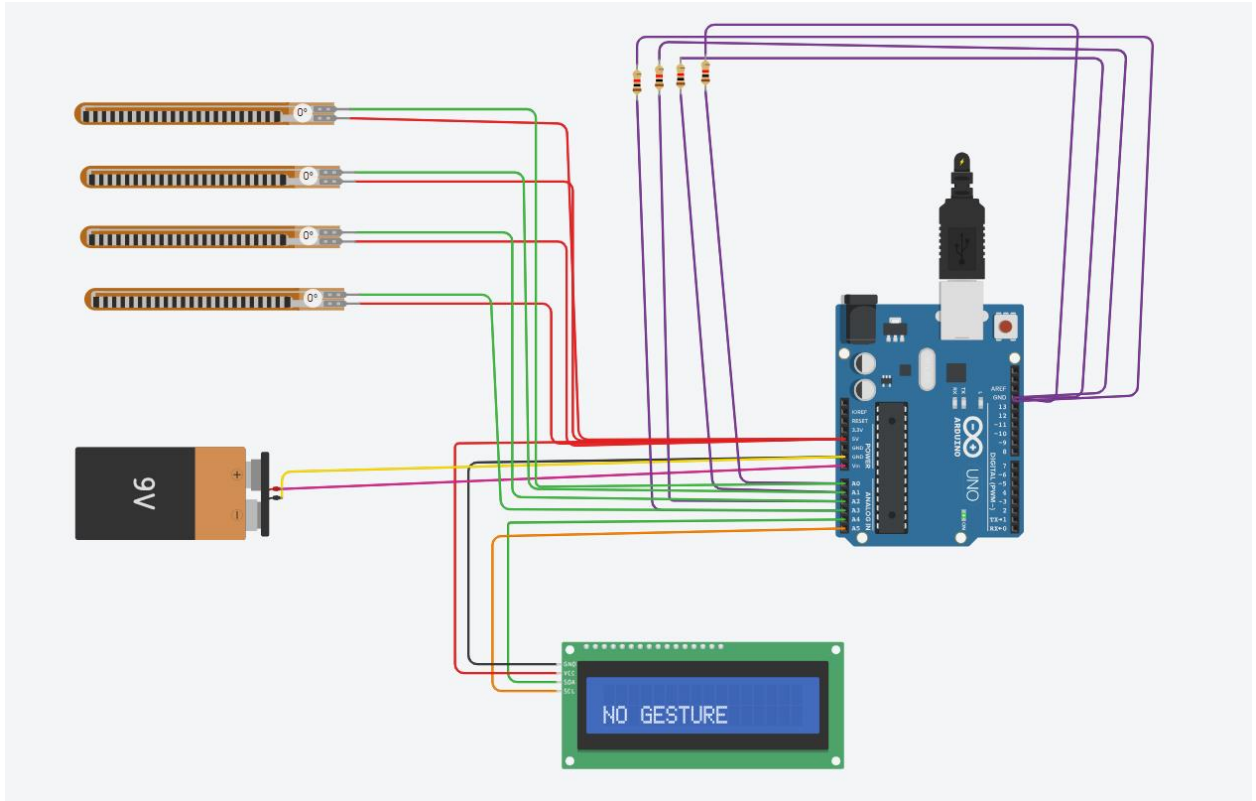
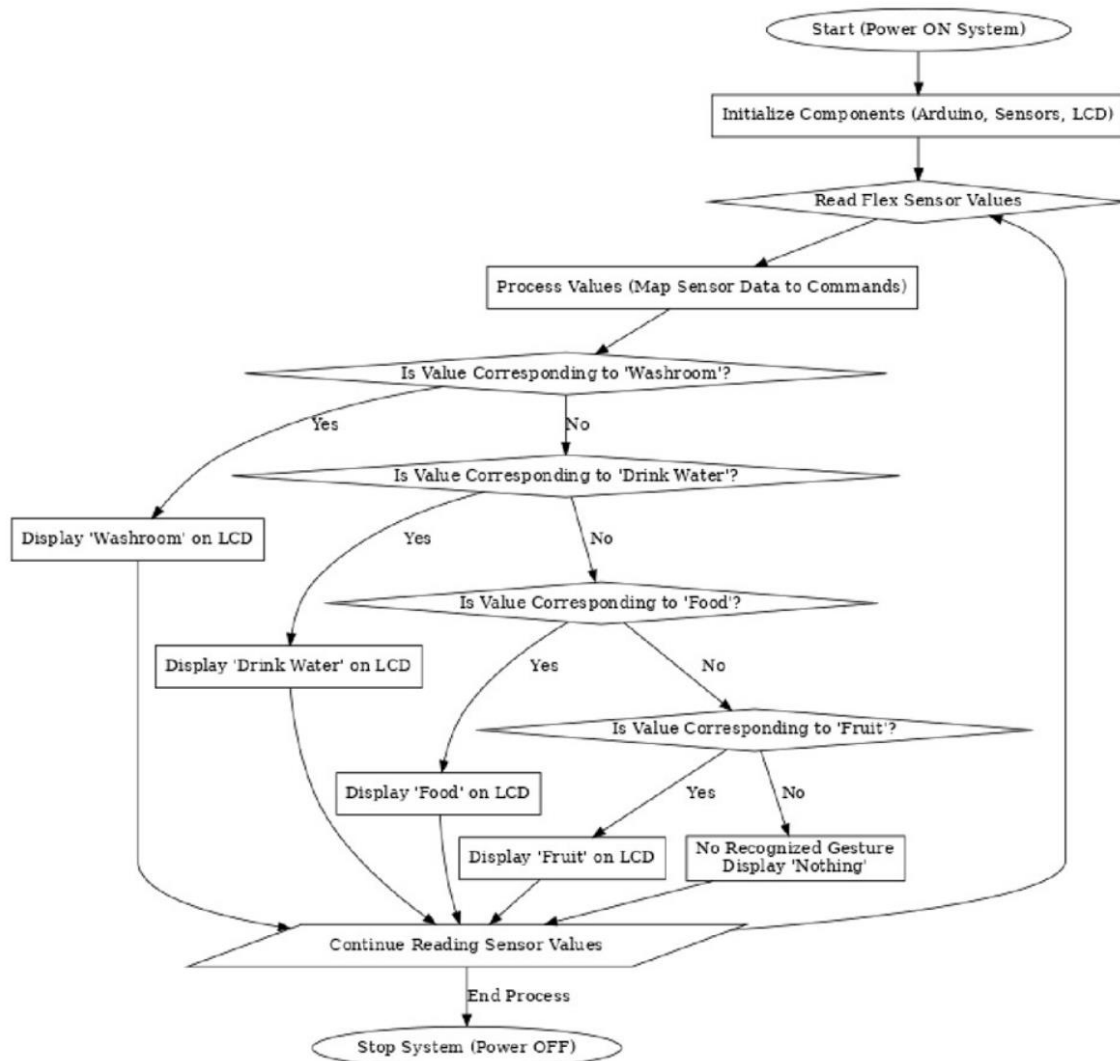


Figure 1: Circuit Diagram of the Model

3.2 Results and Analysis

The glove was tested with predefined gestures for common phrases such as "Washroom," "Drink Water," "Food," and "Fruit." The system demonstrated a 95% accuracy rate in recognizing gestures and correctly displaying corresponding text on the LCD screen. By calibrating the flex sensors and fine-tuning the predefined threshold values, errors were minimized, ensuring consistent results across multiple trials. Further improvements, such as integrating machine learning algorithms, could enhance accuracy by adapting to different users and recognizing a wider range of gestures.



Chapter-IV: Conclusion & Future Scope

4.1 Conclusion

The IoT-based Sign Language Glove successfully translates predefined hand gestures into readable text, addressing a critical communication barrier for individuals with speech and hearing impairments. By utilizing flex sensors, an Arduino Uno R3, and an I2C LCD module, the system provides an effective, real-time solution that is both affordable and portable. The high accuracy of gesture recognition and ease of use make it a promising assistive technology.

One of the key advantages of this system is its portability and independence from external factors such as lighting conditions or complex computational requirements. Unlike vision-based systems, which rely on cameras and advanced image processing, this glove-based solution offers a low-cost, power-efficient alternative that can be used in various real-world scenarios.

Through extensive testing and calibration, the system achieved a high level of accuracy in detecting predefined gestures, making it a reliable communication tool. The integration of an LCD display allows users to receive instant feedback, ensuring effective real-time interaction. Additionally, its simple hardware design and open-source implementation make it accessible to a wide range of users, including students, researchers, and individuals with disabilities.

Despite its effectiveness, there is still room for improvement. The current system is limited to recognizing only a set number of predefined gestures. Expanding its functionality by incorporating machine learning techniques could enhance its ability to recognize a broader range of gestures and improve adaptability to individual users' hand movements. Furthermore, integrating text-to-speech synthesis and wireless communication features would make the system even more versatile and practical in different settings, such as public spaces, workplaces, and educational institutions.

In conclusion, the IoT-based Sign Language Glove serves as a significant step toward breaking communication barriers for individuals with speech and hearing impairments. By continuing to refine and expand this technology, it has the potential to become a widely adopted assistive tool that enhances inclusivity and accessibility for countless users worldwide. The ongoing development of this project will pave the way for more sophisticated and adaptive assistive communication devices in the future.

4.2 Future Scope

The IoT-based Sign Language Glove holds immense potential for future advancements and improvements. By leveraging emerging technologies, the system can be enhanced to provide a more comprehensive, efficient, and user-friendly experience. Below are some key areas of future development:

1. **Expansion of Gesture Database:** The system can be upgraded to recognize a larger variety of gestures, including more complex sign language expressions. This will enable better communication for individuals who rely on sign language daily.
2. **Integration with IoT and Cloud Computing:** By connecting the glove to an IoT platform, users can store, track, and analyze gesture data on cloud-based systems. This would facilitate remote communication and data logging for further improvement in assistive communication technologies.
3. **Mobile Application Development:** A dedicated mobile application can be developed to provide an interface for users to customize gestures, track progress, and translate sign language into text or speech in real-time. The app could also offer multilingual support for wider accessibility.
4. **Text-to-Speech Integration:** To further bridge the communication gap, a voice synthesis module can be added, enabling the recognized gestures to be converted into audible speech. This would significantly help individuals with speech impairments to communicate more effectively in different environments.
5. **Machine Learning and AI Implementation:** Implementing artificial intelligence and machine learning algorithms can improve the system's accuracy by adapting to individual users' hand movements and refining gesture recognition based on different hand sizes and variations.
6. **Wireless Connectivity:** Incorporating Bluetooth or Wi-Fi connectivity would allow the glove to communicate with other devices, such as smartphones, tablets, or computers. This feature would enable real-time translation and remote communication, making the system more flexible and efficient.
7. **Gesture Customization and User Profiles:** Future versions of the glove could include a feature where users can define and store their own custom gestures. This would be especially useful for individuals with unique movement patterns, ensuring a more personalized and adaptive system.

8. **Battery Efficiency and Power Management:** Improving battery life and optimizing power consumption can enhance the device's usability, ensuring prolonged operation without frequent recharging.
9. **Integration with Wearable Technology:** The glove can be combined with smartwatches, augmented reality (AR) glasses, or other wearable devices to provide an immersive communication experience. This integration can also facilitate hands-free operation in various applications.
10. **Haptic Feedback System:** Adding a haptic feedback mechanism can provide tactile responses to users, enhancing interaction and confirming successful gesture recognition.

By implementing these advancements, the IoT-based Sign Language Glove can evolve into a comprehensive assistive communication tool, further bridging the gap between individuals with speech and hearing impairments and the wider community. The incorporation of AI, IoT, and wearable technology will ensure that this innovation remains relevant, scalable, and beneficial for future generations.